

TP — Browser Fingerprinting

Omar Ait Said

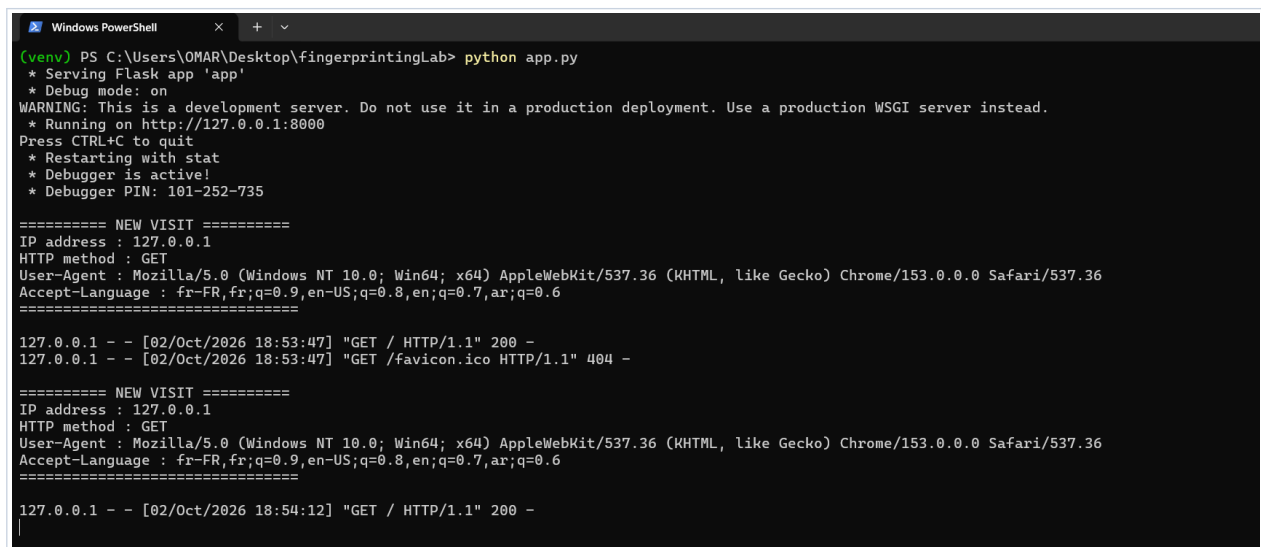
UM6P — College of Computing · Web Privacy · Supervisor: Imane Fouad · 02/10/2026

Part 1 — Investigate the HTTP Request (passive fingerprinting)

1.2 — Project setup

```
fingerprintingLab/  
|-- app.py  
|-- templates/index.html  
|-- static/fingerprint.js  
|-- static/fingerprint2.js
```

```
python -m venv venv  
venv\Scripts\activate  
pip install flask  
python app.py
```



```
Windows PowerShell  
(venv) PS C:\Users\OMAR\Desktop\fingerprintingLab> python app.py  
* Serving Flask app 'app'  
* Debug mode: on  
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.  
* Running on http://127.0.0.1:8000  
Press CTRL+C to quit  
* Restarting with stat  
* Debugger is active!  
* Debugger PIN: 101-252-735  
  
===== NEW VISIT =====  
IP address : 127.0.0.1  
HTTP method : GET  
User-Agent : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36  
Accept-Language : fr-FR,fr;q=0.9,en-US;q=0.8,en;q=0.7,ar;q=0.6  
=====
```

```
127.0.0.1 - - [02/Oct/2026 18:53:47] "GET / HTTP/1.1" 200 -  
127.0.0.1 - - [02/Oct/2026 18:53:47] "GET /favicon.ico HTTP/1.1" 404 -  
  
===== NEW VISIT =====  
IP address : 127.0.0.1  
HTTP method : GET  
User-Agent : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36  
Accept-Language : fr-FR,fr;q=0.9,en-US;q=0.8,en;q=0.7,ar;q=0.6  
=====
```

```
127.0.0.1 - - [02/Oct/2026 18:54:12] "GET / HTTP/1.1" 200 -
```

Figure 1 — Flask server running on 127.0.0.1:8000; each visit prints the IP, method, User-Agent and Accept-Language.

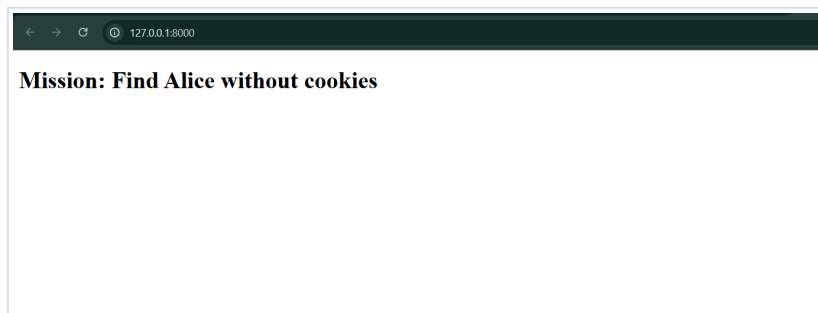


Figure 2 — The page served by Flask (no JavaScript yet).

Task 0 — Investigate the HTTP request

Request headers observed in DevTools (Network → 127.0.0.1 → Headers) and what each one reveals:

Header	Observed value (Chrome)	Reveals	Useful for FP
User-Agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) ... Chrome/153.0.0.0	Browser, version, OS, CPU architecture	Yes
Sec-CH-UA	"Google Chrome";v="153", "Chromium";v="153"	Browser brand and major version	Yes
Sec-CH-UA-Mobile	?0	Mobile or desktop device	Yes
Sec-CH-UA-Platform	"Windows"	Operating system	Yes
Accept	text/html, ..., image/avif, image/webp, ...	Supported content formats (browser-specific)	Yes
Accept-Language	fr-FR,fr;q=0.9,en-US;q=0.8,en;q=0.7,ar;q=0.6	Preferred languages and their order	Yes (high)
Accept-Encoding	gzip, deflate, br, zstd	Supported compression algorithms	Yes (low)
Host	127.0.0.1:8000	Requested server	No
Referer	(absent)	Previous page	No (context)
Sec-Fetch-Site / Mode / User / Dest	cross-site / navigate / ?1 / document	How the request was triggered	No (context)
Connection / Cache-Control	keep-alive / max-age=0	Connection and cache behaviour	No
Upgrade-Insecure-Requests	1	Browser prefers HTTPS	Low
DNT / Sec-GPC	(absent)	Privacy signals; enabling them makes a user rarer	Yes when present
Cookie	(absent)	Stored cookies — none, Alice blocks them	—

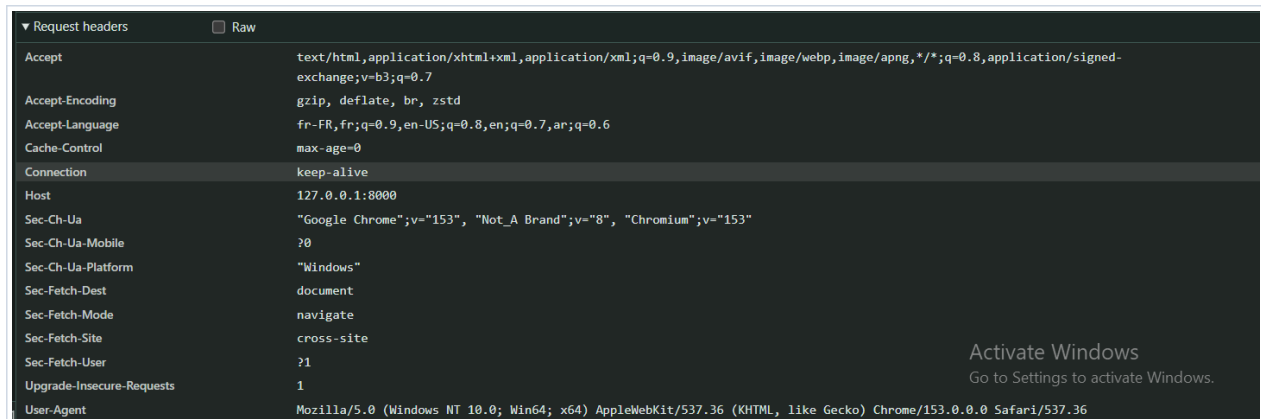


Figure 3 — Request headers of the page request in Chrome DevTools.

Task 1 — Expand the passive collector

1-2. Additional headers identified and printed by app.py, grouped by category:

```
# --- Browser and operating-system information ---
print("Sec-CH-UA           :", request.headers.get("Sec-CH-UA"))
print("Sec-CH-UA-Mobile    :", request.headers.get("Sec-CH-UA-Mobile"))
print("Sec-CH-UA-Platform   :", request.headers.get("Sec-CH-UA-Platform"))
# --- Preferred content formats and languages ---
print("Accept               :", request.headers.get("Accept"))
print("Accept-Encoding       :", request.headers.get("Accept-Encoding"))
# --- Navigation context ---
print("Referer               :", request.headers.get("Referer"))
print("Sec-Fetch-Site         :", request.headers.get("Sec-Fetch-Site"))
print("Sec-Fetch-Mode         :", request.headers.get("Sec-Fetch-Mode"))
print("Sec-Fetch-User         :", request.headers.get("Sec-Fetch-User"))
print("Sec-Fetch-Dest         :", request.headers.get("Sec-Fetch-Dest"))
print("Connection            :", request.headers.get("Connection"))
print("Cache-Control          :", request.headers.get("Cache-Control"))
# --- Security / privacy related headers ---
print("Upgrade-Insecure-Requests :", request.headers.get("Upgrade-Insecure-Requests"))
print("DNT (Do Not Track)       :", request.headers.get("DNT"))
print("Sec-GPC (Global Privacy)  :", request.headers.get("Sec-GPC"))
print("Cookie                  :", request.headers.get("Cookie"))
```

3-4. Observed values in Chrome (normal), Chrome (private window) and Microsoft Edge:

```
(venv) PS C:\Users\OMAR\Desktop\fingerprintingLab> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 101-252-735

===== NEW VISIT =====
IP address       : 127.0.0.1
HTTP method      : GET
HTTP version     : HTTP/1.1
Host             : 127.0.0.1:8000
User-Agent       : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
Sec-CH-UA        : "Google Chrome";v="153", "Not_A Brand";v="8", "Chromium";v="153"
Sec-CH-UA-Mobile : ?0
Sec-CH-UA-Platform : "Windows"
Accept           : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Language  : fr-FR,fr;q=0.9,en-US;q=0.8,en;q=0.7,ar;q=0.6
Accept-Encoding  : gzip, deflate, br, zstd
Referer          : None
Sec-Fetch-Site   : cross-site
Sec-Fetch-Mode   : navigate
Sec-Fetch-User   : ?1
Sec-Fetch-Dest   : document
Connection       : keep-alive
Cache-Control    : None
Upgrade-Insecure-Requests : 1
DNT (Do Not Track) : None
Sec-GPC (Global Privacy) : None
Cookie           : None

=====
127.0.0.1 - - [02/Oct/2026 19:00:24] "GET / HTTP/1.1" 200 -
```

Figure 4 — Passive features collected by the server — Chrome, normal window.

```
===== NEW VISIT =====
IP address       : 127.0.0.1
HTTP method      : GET
HTTP version     : HTTP/1.1
Host             : 127.0.0.1:8000
User-Agent       : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
Sec-CH-UA        : "Google Chrome";v="153", "Not_A Brand";v="8", "Chromium";v="153"
Sec-CH-UA-Mobile : ?0
Sec-CH-UA-Platform : "Windows"
Accept           : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Language  : fr-FR,fr;q=0.9,en-US;q=0.8,en;q=0.7,ar;q=0.6
Accept-Encoding  : gzip, deflate, br, zstd
Referer          : None
Sec-Fetch-Site   : none
Sec-Fetch-Mode   : navigate
Sec-Fetch-User   : ?1
Sec-Fetch-Dest   : document
Connection       : keep-alive
Cache-Control    : None
Upgrade-Insecure-Requests : 1
DNT (Do Not Track) : None
Sec-GPC (Global Privacy) : None
Cookie           : None

=====
127.0.0.1 - - [02/Oct/2026 19:01:35] "GET / HTTP/1.1" 200 -
```

Figure 5 — Passive features — Chrome, private (Incognito) window.

```

===== NEW VISIT =====
IP address      : 127.0.0.1
HTTP method     : GET
HTTP version    : HTTP/1.1
Host            : 127.0.0.1:8800
User-Agent      : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36 Edg/154.0.0
Sec-CH-UA       : "Chromium";v="154", "Microsoft Edge";v="154", "Not A(Brand";v="99"
Sec-CH-UA-Mobile : ?0
Sec-CH-UA-Platform : "Windows"
Accept          : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Language : en-US,en;q=0.9
Accept-Encoding : gzip, deflate, br, zstd
Referer         : None
Sec-Fetch-Site  : none
Sec-Fetch-Mode  : navigate
Sec-Fetch-User  : ?1
Sec-Fetch-Dest  : document
Connection      : keep-alive
Cache-Control   : None
Upgrade-Insecure-Requests : 1
DNT (Do Not Track) : None
Sec-GPC (Global Privacy) : None
Cookie          : None
=====

127.0.0.1 - - [02/Oct/2026 19:02:31] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 19:02:31] "GET /favicon.ico HTTP/1.1" 404 -

```

Figure 6 — Passive features — Microsoft Edge.

5. Values that remain the same and values that change:

Header	Chrome	Chrome private	Edge	Result
User-Agent	Chrome/153	Chrome/153	Chrome/154 ... Edg/154	Stable in private, changes with browser
Sec-CH-UA	Google Chrome 153	Google Chrome 153	Microsoft Edge 154	Stable in private, changes with browser
Sec-CH-UA-Platform	"Windows"	"Windows"	"Windows"	Stable (same OS)
Accept	identical	identical	identical	Stable (same engine)
Accept-Language	fr-FR,fr,en-US,en,ar	fr-FR,fr,en-US,en,ar	en-US,en	Stable in private, changes with browser
Accept-Encoding	gzip, deflate, br, zstd	same	same	Stable
Sec-Fetch-Site	cross-site	none	none	Changes with navigation context
Cookie / DNT / Sec-GPC	None	None	None	Not sent

Private browsing does not change the passive fingerprint: only the navigation-context headers vary. Changing browser changes the User-Agent, Client Hints and Accept-Language.

Part 2 — Active Feature Collection

2.2 / 2.3 — First active feature

```

<!-- templates/index.html, before </body> -->
<h2>Active Feature Collection</h2>
<div id="feature-output">Waiting for JavaScript...</div>
<script src="{ url_for('static', filename='fingerprint.js') }}"></script>

// static/fingerprint.js
const browserLanguage = navigator.language;
console.log("Browser language:", browserLanguage);
const outputElement = document.getElementById("feature-output");
outputElement.textContent = "Browser language: " + browserLanguage;

```

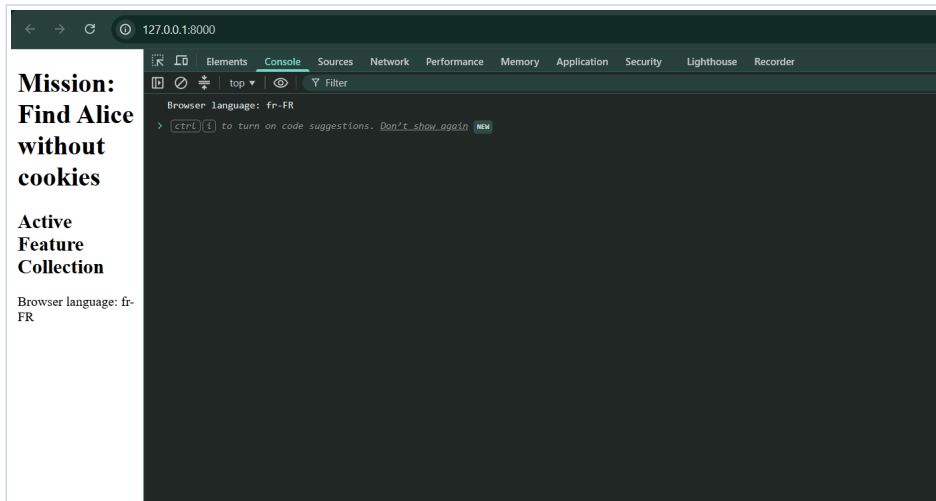


Figure 7 — `navigator.language` read by JavaScript and displayed on the page and in the console.

Task 2 — Expand the active collector

1. Collected features (from navigator, screen, window, Intl, Date and document):

```
function collectActiveFeatures() {
  return {
    language: navigator.language,
    languages: (navigator.languages || []).join(","),
    userAgent: navigator.userAgent,
    platform: navigator.platform,
    hardwareConcurrency: navigator.hardwareConcurrency,
    deviceMemory: navigator.deviceMemory !== undefined ? navigator.deviceMemory : "not exposed",
    cookieEnabled: navigator.cookieEnabled,
    doNotTrack: navigator.doNotTrack,
    screenResolution: screen.width + "x" + screen.height,
    availableScreen: screen.availWidth + "x" + screen.availHeight,
    colorDepth: screen.colorDepth,
    windowSize: window.innerWidth + "x" + window.innerHeight,
    devicePixelRatio: window.devicePixelRatio,
    timeZone: Intl.DateTimeFormat().resolvedOptions().timeZone,
    locale: Intl.DateTimeFormat().resolvedOptions().locale,
    timezoneOffset: new Date().getTimezoneOffset(),
    referrer: document.referrer || "(none)",
    visibilityState: document.visibilityState
  };
}
```

2. Display on the page, 3. send to Flask:

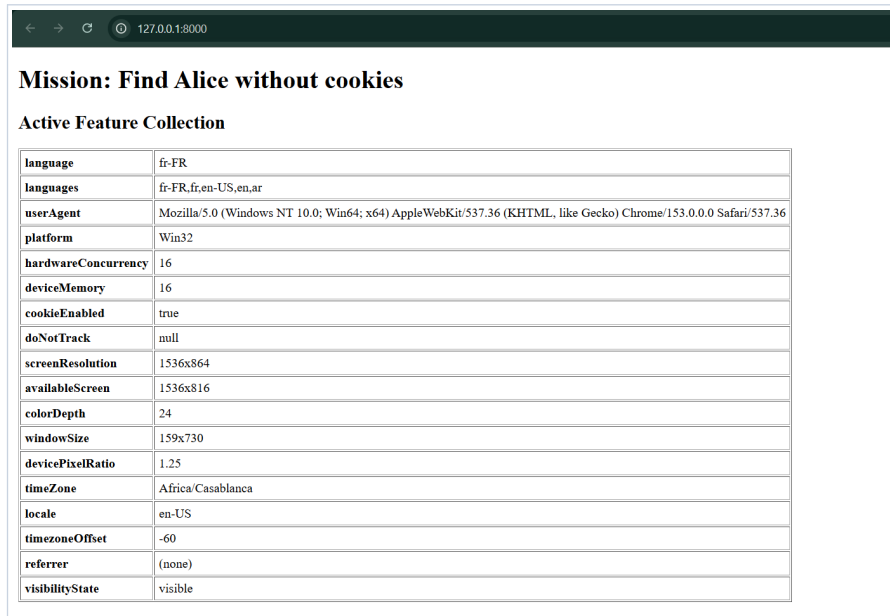
```
function displayFeatures(features) {
  let html = "<table border='1' cellpadding='5'>";
  for (const name in features)
    html += "<tr><td><b>" + name + "</b></td><td>" + features[name] + "</td></tr>";
  document.getElementById("feature-output").innerHTML = html + "</table>";
}

function sendFeatures(features) {
  fetch("/collect", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ type: "active-features", features: features })
  }).then(r => r.json()).then(a => console.log("Server answer:", a));
}

const activeFeatures = collectActiveFeatures();
displayFeatures(activeFeatures);
sendFeatures(activeFeatures);
```

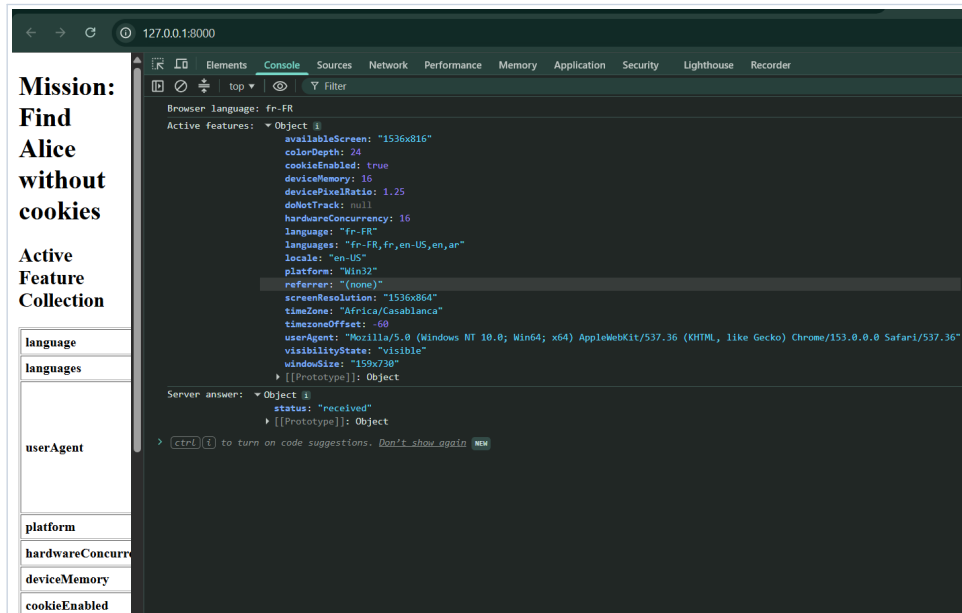
4. Flask route receiving and printing the features:

```
@app.route("/collect", methods=["GET", "POST"])
def collect():
    print("\n===== DATA RECEIVED ON /collect =====")
    print("IP address :", request.remote_addr)
    data = request.get_json(silent=True)
    if data:
        print("Data type :", data.get("type"))
        for key, value in data.get("features", {}).items():
            print(f"  {key:<22}: {value}")
    return jsonify({"status": "received"})
```



language	fr-FR
languages	fr-FR,fr,en-US,en,ar
userAgent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
platform	Win32
hardwareConcurrency	16
deviceMemory	16
cookieEnabled	true
doNotTrack	null
screenResolution	1536x864
availableScreen	1536x816
colorDepth	24
windowSize	159x730
devicePixelRatio	1.25
timeZone	Africa/Casablanca
locale	en-US
timezoneOffset	-60
referrer	(none)
visibilityState	visible

Figure 8 — Active features displayed on the webpage (Chrome).



Mission: Find Alice without cookies

Active Feature Collection

language	fr-FR
languages	fr-FR,fr,en-US,en,ar
userAgent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
platform	Win32
hardwareConcurrency	16
deviceMemory	16
cookieEnabled	true

Console:

```
Browser language: fr-FR
Active features: Object {
  availableScreen: "1536x816"
  colorDepth: 24
  cookieEnabled: true
  deviceMemory: 16
  devicePixelRatio: 1.25
  doNotTrack: null
  hardwareConcurrency: 16
  language: "fr-FR"
  languages: "fr-FR,fr,en-US,en,ar"
  locale: "en-US"
  platform: "Win32"
  referrer: "(none)"
  screenResolution: "1536x864"
  timeZone: "Africa/Casablanca"
  timezoneOffset: -60
  userAgent: "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
  visibilityState: "visible"
  windowSize: "159x730"
}
Server answer: Object {
  status: "received"
}
```

Figure 9 — Console: collected features and the server's answer {status: "received"}.

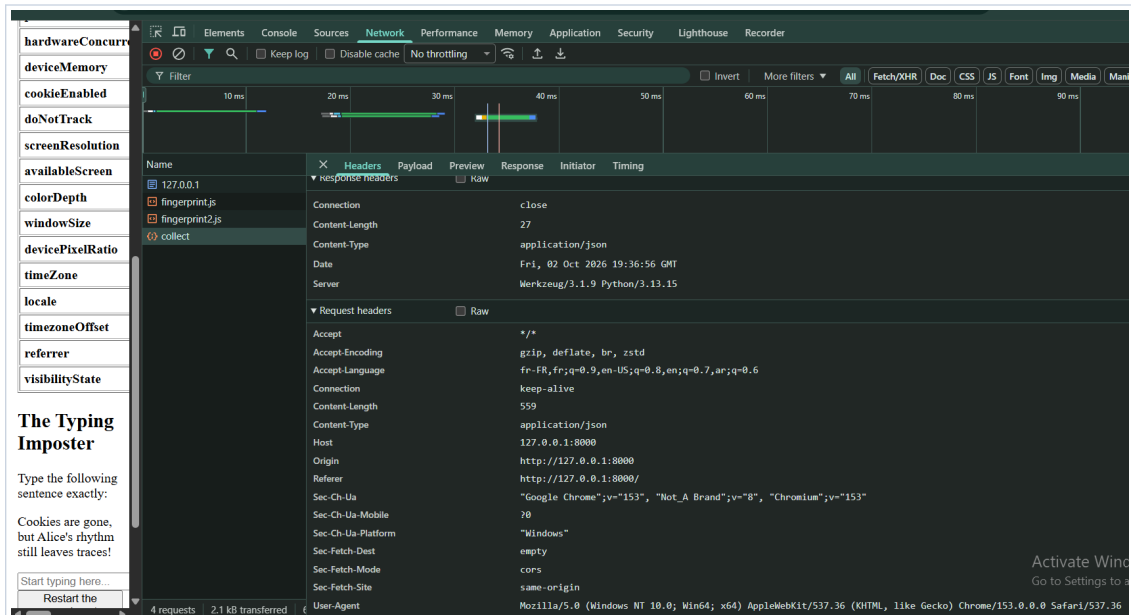


Figure 10 — DevTools Network: POST request to /collect carrying the features as JSON (Content-Type: application/json).

```

===== DATA RECEIVED ON /collect =====
IP address : 127.0.0.1
Data type : active-features
language      : fr-FR
languages     : fr-FR,fr,en-US,en,ar
userAgent     : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
platform      : Win32
hardwareConcurrency : 16
deviceMemory  : 16
cookieEnabled : True
doNotTrack    : None
screenResolution : 1536x864
availableScreen : 1536x816
colorDepth    : 24
windowSize    : 159x730
devicePixelRatio : 1.25
timeZone      : Africa/Casablanca
locale        : en-US
timezoneOffset : -60
referrer      : (none)
visibilityState : visible
=====

127.0.0.1 - - [02/Oct/2026 19:07:54] "POST /collect HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 19:08:24] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -

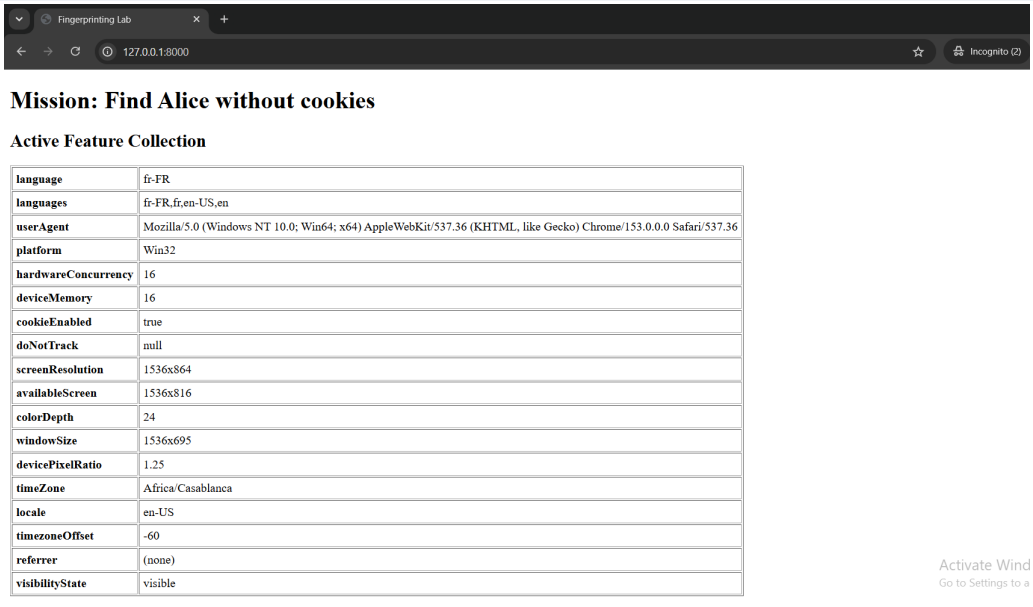
```

Figure 11 — Flask console printing the active features received on /collect.

What each feature reveals and why it distinguishes browsers:

Feature	Object	My value	What it reveals / why it is discriminating
languages	navigator	fr-FR,fr,en-US,en,ar	Ordered language list; the combination French + English + Arabic is rare
hardwareConcurrency	navigator	16	Logical CPU cores; varies between machines (4, 8, 12, 16...)
deviceMemory	navigator	16	Approximate RAM in GB; separates low-end and high-end devices
userAgent / platform	navigator	Chrome 153 / Win32	Browser, version and OS
screenResolution	screen	1536x864	Physical screen size; depends on the device
availableScreen	screen	1536x816	Screen minus taskbar; reveals OS layout settings
colorDepth	screen	24	Colour depth; distinguishes HDR screens
devicePixelRatio	window	1.25	Windows display scaling (125 %); varies per screen and settings
windowSize	window	varies	Current window size; unstable (changes on resize)
timeZone / timezoneOffset	Intl / Date	Africa/Casablanca / -60	Geographic region, even behind a VPN
referrer / visibilityState	document	(none) / visible	Navigation context; not stable

Task 2.5 — Protection mechanisms



The screenshot shows a Chrome Incognito window titled 'Fingerprinting Lab'. The address bar shows '127.0.0.1:8000'. The page content includes a mission statement and a table of active browser features.

Mission: Find Alice without cookies

Active Feature Collection

language	fr-FR
languages	fr-FR,fr,en-US,en
userAgent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
platform	Win32
hardwareConcurrency	16
deviceMemory	16
cookieEnabled	true
doNotTrack	null
screenResolution	1536x864
availableScreen	1536x816
colorDepth	24
windowSize	1536x695
devicePixelRatio	1.25
timeZone	Africa/Casablanca
locale	en-US
timezoneOffset	-60
referrer	(none)
visibilityState	visible

Activate Wind
Go to Settings to a

Figure 12 — Active features in a Chrome private (Incognito) window.

```

===== DATA RECEIVED ON /collect =====
IP address : 127.0.0.1
Data type : active-features
language      : fr-FR
languages     : fr-FR,fr,en-US,en
userAgent     : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
platform      : Win32
hardwareConcurrency : 16
deviceMemory  : 16
cookieEnabled : True
doNotTrack    : None
screenResolution : 1536x864
availableScreen : 1536x816
colorDepth    : 24
windowSize    : 1536x695
devicePixelRatio : 1.25
timeZone      : Africa/Casablanca
locale        : en-US
timeZoneOffset : -60
referrer      : (none)
visibilityState : visible
=====
127.0.0.1 - - [02/Oct/2026 19:11:40] "POST /collect HTTP/1.1" 200 -

```

Figure 13 — Flask console — features received from the private window.

Fingerprinting Lab

127.0.0.1:8000

Mission: Find Alice without cookies

Active Feature Collection

language	en-US
languages	en-US,en
userAgent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36 Edg/154.0.0.0
platform	Win32
hardwareConcurrency	16
deviceMemory	16
cookieEnabled	true
doNotTrack	null
screenResolution	1536x864
availableScreen	1536x816
colorDepth	24
windowSize	1528x732
devicePixelRatio	1.25
timeZone	Africa/Casablanca
locale	en-US
timeZoneOffset	0
referrer	(none)
visibilityState	visible

Figure 14 — Active features in Microsoft Edge.

```

===== DATA RECEIVED ON /collect =====
IP address : 127.0.0.1
Data type : active-features
language      : en-US
languages     : en-US,en
userAgent     : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36 Edg/154.0.0.0
platform      : Win32
hardwareConcurrency : 16
deviceMemory  : 16
cookieEnabled : True
doNotTrack    : None
screenResolution : 1536x864
availableScreen : 1536x816
colorDepth    : 24
windowSize    : 1528x732
devicePixelRatio : 1.25
timeZone      : Africa/Casablanca
locale        : en-US
timeZoneOffset : 0
referrer      : (none)
visibilityState : visible
=====
127.0.0.1 - - [02/Oct/2026 19:28:23] "POST /collect HTTP/1.1" 200 -

```

Figure 15 — Flask console — features received from Edge.

Feature	Chrome	Chrome private	Edge	Stable?
language	fr-FR	fr-FR	en-US	Changes with browser
languages	fr-FR,fr,en-US,en,ar	fr-FR,fr,en-US,en	en-US,en	Reduced in private mode
userAgent	Chrome/153	Chrome/153	Edg/154	Changes with browser
hardwareConcurrency	16	16	16	Stable
deviceMemory	16	16	16	Stable

Feature	Chrome	Chrome private	Edge	Stable?
screen / available	1536x864 / 1536x816	same	same	Stable
colorDepth / DPR	24 / 1.25	same	same	Stable
windowSize	159x730	1536x695	1528x732	Unstable
timeZone	Africa/Casablanca	same	same	Stable
timezoneOffset	-60	-60	0	Changes with browser

Protection	Effect on the fingerprint
Private browsing	Hardware, screen, time zone and User-Agent unchanged; only the language list is reduced. Not effective.
Cookie blocking	No collected feature is stored in the browser, so the fingerprint is unaffected. Not effective.
Another browser	Changes User-Agent, languages and offset, but hardware, screen and time zone stay identical.
Brave (randomisation)	Randomises some values (e.g. hardwareConcurrency) per session and per site. Partially effective.
Tor Browser / Firefox resistFingerprinting	Standardises values for all users: UTC time zone, en-US, common core count, window-sized screen, DPR 1. Most effective.

Most effective protection: standardisation (Tor Browser or Firefox privacy.resistFingerprinting), because every user reports the same values and the fingerprint blends into a crowd.

Limitations: it breaks some websites (wrong time zone and language); the small number of users of these protections makes them identifiable as a group; the IP address and passive headers remain; it does not protect against behavioural fingerprinting. Randomisation can be averaged over several visits and its inconsistency can be detected.

Part 3 — The Typing Imposter

Task 3 — The HTML interface

```
<h2>The Typing Imposter</h2>
<p>Type the following sentence exactly:</p>
<p id="target-sentence">Cookies are gone, but Alice's rhythm still leaves traces!</p>
<input type="text" id="typing-input" size="70" placeholder="Start typing here..."
  autocomplete="off" autocorrect="off" autocapitalize="off" spellcheck="false">
<button id="restart-btn">Restart the experiment</button>
<div id="typing-results">
  <p>Total typing time: <span id="result-time"></span></p>
  <p>Typing speed: <span id="result-speed"></span></p>
  <p>Number of corrections (Backspace): <span id="result-corrections"></span></p>
</div>
<script src="{ url_for('static', filename='fingerprint2.js') }"></script>
```

hardwareConcurrency	16
deviceMemory	16
cookieEnabled	true
doNotTrack	null
screenResolution	1536x864
availableScreen	1536x816
colorDepth	24
windowSize	1536x730
devicePixelRatio	1.25
timeZone	Africa/Casablanca
locale	en-US
timezoneOffset	-60
referrer	(none)
visibilityState	visible

The Typing Imposter

Type the following sentence exactly:

Cookies are gone, but Alice's rhythm still leaves traces!

Start typing here...

Total typing time:

Typing speed:

Number of corrections (Backspace):

Figure 16 — Typing interface before the experiment: sentence, input, restart button, empty result area.

Task 4 — Alice's typing rhythm

1. Formula

Measure	Formula
Typing speed (characters/s)	number of characters of the sentence ÷ total time in seconds
Words per minute (WPM)	(number of characters ÷ 5) ÷ total time in minutes (1 word = 5 characters)

3. Algorithm

1. Get the sentence, the input, the button and the result areas with `getElementById()`.
2. Initialise `startTime = null`, `corrections = 0`, `finished = false`.
3. On **keydown**: if `startTime` is null and a character key is pressed → `startTime = performance.now()`; if the key is Backspace or Delete → `corrections + 1`.
4. On **input**: if `input.value === target sentence` → `endTime = performance.now()`, `finished = true`.
5. Compute `time = (endTime - startTime) / 1000`, speed and WPM; display them.
6. Put the results in an object and send it to `/collect` with `fetch()` (JSON).
7. On **click** of the restart button: reset all variables, the input and the results.

4-5. static/fingerprint2.js

```
const targetElement = document.getElementById("target-sentence");
const typingInput = document.getElementById("typing-input");
const restartButton = document.getElementById("restart-btn");
const timeOutput = document.getElementById("result-time");
const speedOutput = document.getElementById("result-speed");
const correctionsOutput = document.getElementById("result-corrections");
const targetSentence = targetElement.textContent.trim();

let startTime = null, corrections = 0, finished = false;

typingInput.addEventListener("keydown", function (event) {
  if (finished) { event.preventDefault(); return; }
  if (startTime === null && event.key.length === 1) startTime = performance.now();
  if ((event.key === "Backspace" || event.key === "Delete") && startTime !== null) corrections++;
});

typingInput.addEventListener("input", function () {
  if (!finished && startTime !== null && typingInput.value === targetSentence) {
    const endTime = performance.now();
    finished = true;
    const totalSeconds = (endTime - startTime) / 1000;
    const charsPerSecond = targetSentence.length / totalSeconds;
    const wpm = (targetSentence.length / 5) / (totalSeconds / 60);

    timeOutput.textContent = totalSeconds.toFixed(2) + " s";
    speedOutput.textContent = charsPerSecond.toFixed(2) + " characters/s (" + wpm.toFixed(2) + " WPM)";
    correctionsOutput.textContent = corrections;

    const typingFeatures = {
      typingTime: Number(totalSeconds.toFixed(2)),
      charsPerSecond: Number(charsPerSecond.toFixed(2)),
      wordsPerMinute: Number(wpm.toFixed(2)),
      corrections: corrections
    };
    fetch("/collect", {
      method: "POST",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify({ type: "typing-behaviour", features: typingFeatures })
    });
  }
});

restartButton.addEventListener("click", function () {
  startTime = null; corrections = 0; finished = false;
  typingInput.value = "";
  timeOutput.textContent = speedOutput.textContent = correctionsOutput.textContent = "";
  typingInput.focus();
});
```

6. Results and verification:

deviceMemory	16
cookieEnabled	true
doNotTrack	null
screenResolution	1536x864
availableScreen	1536x816
colorDepth	24
windowSize	1536x730
devicePixelRatio	1.25
timeZone	Africa/Casablanca
locale	en-US
timezoneOffset	-60
referrer	(none)
visibilityState	visible

The Typing Imposter

Type the following sentence exactly:

Cookies are gone, but Alice's rhythm still leaves traces!

Cookies are gone, but Alice's rhythm still leaves traces!

Total typing time: 11.62 s

Typing speed: 4.90 characters/s (58.85 WPM)

Number of corrections (Backspace): 0

Figure 17 — Trial 1 — sentence completed in 11.62 s, 4.90 characters/s (58.85 WPM), 0 corrections.

```

===== DATA RECEIVED ON /collect =====
IP address : 127.0.0.1
Data type : typing-behaviour
typingTime      : 11.62
charsPerSecond  : 4.9
wordsPerMinute  : 58.85
corrections     : 0
=====

127.0.0.1 - - [02/Oct/2026 19:34:26] "POST /collect HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 19:35:06] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -

```

Figure 18 — Flask console — typing features of trial 1 received on /collect.

The Typing Imposter

Type the following sentence exactly:

Cookies are gone, but Alice's rhythm still leaves traces!

Total typing time: 31.22 s

Typing speed: 1.83 characters/s (21.91 WPM)

Number of corrections (Backspace): 3

Figure 19 — Trial 2 — 31.22 s, 1.83 characters/s (21.91 WPM), 3 Backspace corrections.

```

===== DATA RECEIVED ON /collect =====
IP address : 127.0.0.1
Data type : typing-behaviour
typingTime      : 31.22
charsPerSecond  : 1.83
wordsPerMinute  : 21.91
corrections     : 3
=====

127.0.0.1 - - [02/Oct/2026 19:42:33] "POST /collect HTTP/1.1" 200 -

```

Figure 20 — Flask console — typing features of trial 2 received on /collect.

Trial	Time (s)	Speed (char/s)	WPM	Corrections
1	11.62	4.90	58.85	0
2	31.22	1.83	21.91	3

The same user produced different values in each trial: typing behaviour is probabilistic and must be compared with a tolerance over many samples, not as an exact identifier.